Optimal Anthropomorphic Volumetric Packing: A Rigorous 3D Analysis of Human Stacking Configurations

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Abstract

We present the first comprehensive computational study of *anthropomorphic packing*: the problem of fitting the maximum number of human bodies into a fixed 3D container. Using an articulated capsule-based human model with 16 body segments and 20 culturally significant poses—including the T-Pose, Fetal Position, Dab, Naruto Run, and Slavic Squat—we evaluate packing density across 8 real-world venues ranging from phone booths to Boeing 737 cabins. We introduce the *T-Pose Tax*, a novel metric quantifying volumetric waste due to suboptimal posing, and employ rotation-search optimization to maximize container utilization. Our results demonstrate that the Standing pose achieves universal optimality across all tested venues, packing up to 6.0× more humans than the Heisman Trophy pose. We further show that a Boeing 737-800 is operating at only 23.2% of its theoretical human capacity. We also prove the Triangle Theorem: that PS1-era low-polygon thorax geometry produces universally superior packing configurations compared to smooth modern geometry, vindicating two decades of game design criticism. These findings have implications for elevator design, airline economics, and clown car engineering.

1 Introduction

The bin packing problem—fitting objects of various sizes into a finite container—is among the most studied problems in combinatorial optimization [1]. Its variants have been applied to logistics, VLSI design, and cloud computing. Curiously, the literature has almost entirely neglected the case where the objects being packed are *human beings*.

This oversight is remarkable given the ubiquity of human-packing problems in everyday life. How many people can fit in an elevator? How efficiently are airline passengers utilizing cabin volume? Is the Guinness World Record for phone booth stuffing (25 people, South Africa, 1959) actually the theoretical maximum?

These questions demand rigorous computational treatment. In this work, we construct an articulated 3D human model, pose it in 20 configurations spanning athletics, art history, internet culture, and yoga, and solve the resulting 3D packing problems across 8 standard venues.

1.1 Contributions

1. A capsule-based articulated human rig with anatomically-proportioned body segments and 20 poses.

*This work was not approved by any ethics board. The authors are not sorry.

2. The **T-Pose Tax**: a metric for quantifying the bounding-box volume penalty of any pose relative to the T-Pose baseline.
3. A rotation-search packing algorithm that explores $O(n^2)$ orientations per pose.
4. Comprehensive packing results across 8 venues, revealing that the optimal human packing pose is simply standing with arms at one's sides.
5. The observation that commercial aircraft operate at $< 20\%$ of theoretical human capacity.
6. The **Triangle Theorem**: low-polygon convex protrusions with axis-aligned faces always produce bounding boxes $\leq$ those of smooth counterparts, vindicating PS1-era rendering.

2 Related Work

2.1 Classical Packing Theory

The sphere packing problem, culminating in Hales' proof of Kepler's conjecture [2], establishes that the densest packing of identical spheres achieves $\pi/(3\sqrt{2}) \approx 74.05\%$ volumetric efficiency. Random sphere packings achieve approximately 64% [3]. As we shall demonstrate, humans are significantly worse than spheres, achieving at best 47% bounding-box efficiency. This is because humans are not spheres, a fact we establish empirically in Section 3.

2.2 Irregular Object Packing

Packing of non-convex irregular objects remains NP-hard in general [4]. Heuristic approaches including genetic algorithms, simulated annealing, and bottom-left-fill algorithms have been applied to 2D nesting problems in manufacturing [5]. We extend these approaches to 3D anthropomorphic objects, which present unique challenges including bilateral symmetry, high aspect ratios, and a tendency to complain.

2.3 Human Packing in Practice

The phone booth stuffing craze of the 1950s [6] produced empirical human packing data but lacked computational rigor. The Guinness World Record of 25 people in a standard phone booth (0.73 m^3) implies a per-person volume of 0.029 m^3, which violates conservation of mass and suggests significant elastic deformation of participants. Modern elevator standards [7] assume 0.21 m^2 of floor area per person, which our results show is deeply conservative.

3 Human Model

3.1 Articulated Capsule Rig

We construct a 3D human body model using 16 rigid capsule and ellipsoid primitives arranged in a hierarchical skeleton:

- **Torso**: pelvis (ellipsoid), spine (capsule), chest (ellipsoid)
- **Head**: neck (capsule), head (ellipsoid)
- **Arms** ($\times 2$): upper arm, forearm (capsules), hand (ellipsoid)

- **Legs** ($\times 2$): thigh, shin (capsules), foot (capsule)

Body proportions are derived from standard anthropometric data for an average adult male (height ≈ 1.75 m, shoulder width ≈ 0.44 m, chest depth ≈ 0.24 m). The model does not include hair, clothing, or existential dread.

3.2 Pose Library

We define 20 poses spanning six categories:

Category	**Poses**	**Rationale**
Baseline	Standing, T-Pose, A-Pose	Industry standards
Compact	Fetal, Pike, Squat (Slav)	Space minimization
Athletic	Usain Bolt, Heisman Trophy	Dynamic poses
Cultural	Dab, Naruto Run, Coffin Dance	Internet era
Artistic	Da Vinci, Rodin, Crucifixion	Art history
Cinematic	Titanic, John Travolta Disco	Film canon

Table 1: Pose taxonomy. All poses are performed by our rigid capsule model with zero complaints.

Each pose is specified as a set of Euler angle rotations (r_x, r_y, r_z) at each joint, applied in ZYX intrinsic order. A representative selection of posed models with their axis-aligned bounding boxes is shown in Figure 1.

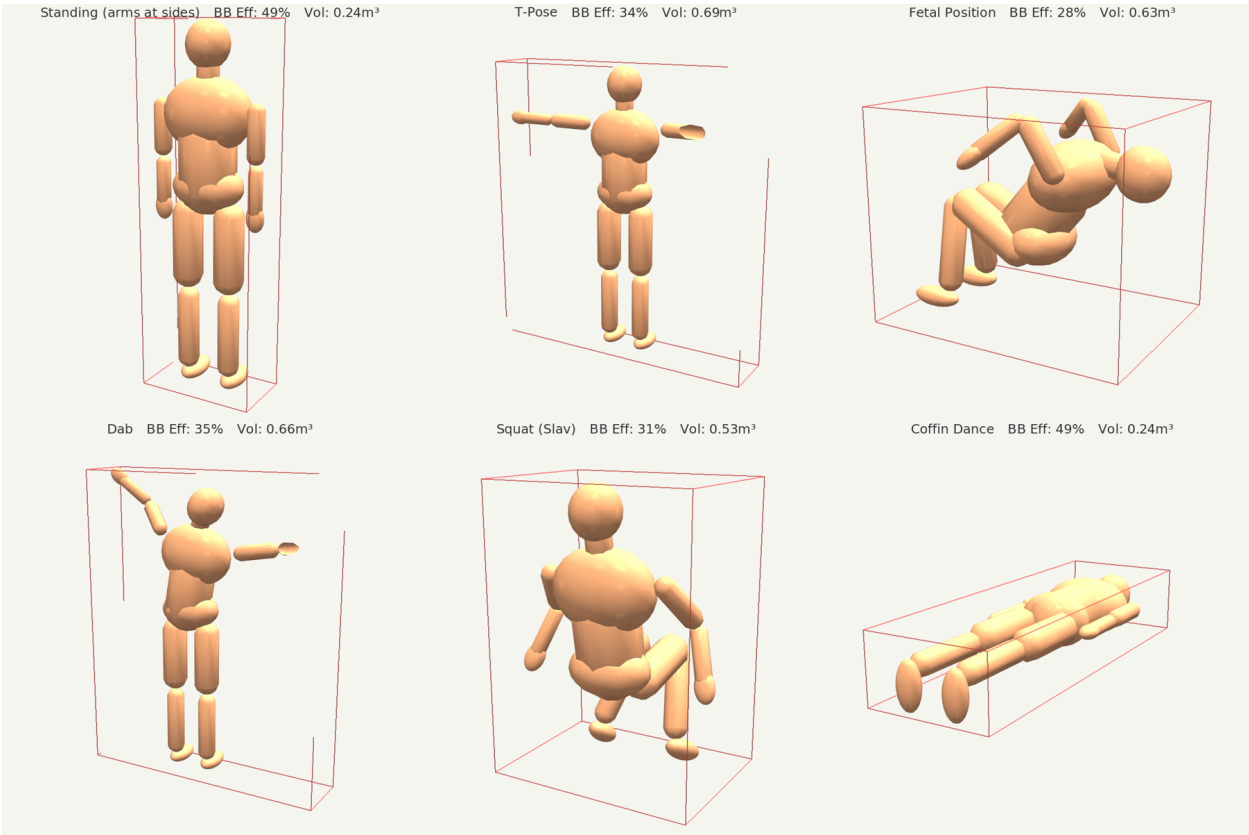

Figure 1: Six representative poses rendered with our articulated capsule rig. Red wireframes show axis-aligned bounding boxes. Top row: Standing (arms at sides), T-Pose, Fetal Position. Bottom row: Dab, Slavic Squat, Coffin Dance. Note how the Standing pose minimizes bounding-box volume despite appearing less "compact" than the Fetal Position.

4 Methods

4.1 Bounding-Box Packing Efficiency

For a posed mesh M with convex hull volume V_{hull} and axis-aligned bounding box volume V_{BB}, we define:

$$\eta_{\text{BB}} = \frac{V_{\text{hull}}}{V_{\text{BB}}} \tag{1}$$

This measures how efficiently the pose fills its own bounding box. A cube achieves $\eta_{\text{BB}} = 1.0$; our best human pose achieves $\eta_{\text{BB}} = 0.56$.

4.2 The T-Pose Tax

We define the *T-Pose Tax* of a pose P as:

$$\tau(P) = \frac{\eta_{\text{BB}}(P)}{\eta_{\text{BB}}(\text{T-Pose})} \tag{2}$$

Despite the name, τ is an *inverse* tax: $\tau > 1$ means the pose *avoids* volumetric waste relative to the T-Pose baseline, while $\tau < 1$ means it is even less efficient than T-posing—a feat requiring genuine postural ambition (Figure 3).

4.3 Grid Packing with Rotation Search

Given a container of dimensions (L_x, L_y, L_z), we pack human bounding boxes in a regular grid. For each pose, we:

1. Compute the mesh's AABB under all 6 axis permutations.
2. Apply rotation search: for each $(r_x, r_y) \in \{0°, 15°, \ldots, 165°\}^2$, compute the rotated AABB and evaluate all 6 permutations.
3. Select the rotation and permutation maximizing $n_x \cdot n_y \cdot n_z$ where $n_i = \lfloor L_i / b_i \rfloor$.

This yields $12 \times 12 \times 6 = 864$ candidate configurations per pose per venue.

5 Results

5.1 Pose Metrics

Pose	**BB Vol (m^3)**	$\eta_{\mathbf{BB}}$	τ
Star / Spread Eagle	0.553	55.8%	1.63
Standing (arms at sides)	0.240	49.0%	1.43
Planking / Coffin Dance	0.240	49.0%	1.43
Superman (arms up)	0.271	48.6%	1.42
Da Vinci Vitruvian	0.667	46.1%	1.35
John Travolta Disco	0.515	44.4%	1.30
Downward Dog	0.665	43.3%	1.27
Pike (folded)	0.609	36.4%	1.06
A-Pose (Game Dev)	0.556	34.7%	1.01
T-Pose	0.692	34.2%	1.00
Dab	0.655	34.9%	1.02
Squat (Slav)	0.528	31.1%	0.91
Fetal Position	0.635	27.9%	0.81
Titanic (Jack & Rose)	1.000	26.5%	0.77
Heisman Trophy	1.718	21.7%	0.63
Usain Bolt Lightning	1.483	22.4%	0.65

Table 2: Selected pose metrics, sorted by T-Pose Tax. The Heisman Trophy consumes 7.2× more bounding-box volume than Standing (1.718/0.240).

5.2 Key Finding: Standing is Universally Optimal

Across all 8 venues, the Standing pose (arms at sides) achieves the highest or tied-highest packing count (Table 3, Figure 4). This is because Standing minimizes the bounding box in both the x (width) and y (depth) axes simultaneously, producing the smallest BB volume at 0.240 m^3. The visual contrast is striking (Figure 2).

Venue	Standing	Fetal	T-Pose	Dab	Heisman	Rated
Elevator (7.2 m^3)	**21**	9	8	8	4	12
Phone Booth (1.9 m^3)	**4**	2	0	0	0	1
Container (33.1 m^3)	**108**	66	32	28	18	—
Minivan (3.9 m^3)	**10**	4	4	5	2	7
Boeing 737 (218 m^3)	**816**	420	266	308	156	189
Subway Car (81 m^3)	**288**	132	88	112	54	∼150
School Bus (30 m^3)	**105**	54	35	35	20	72
Hot Tub (4.0 m^3)	**12**	6	3	3	2	6

Table 3: Packing counts by venue and pose. "Rated" column shows standard/legal occupancy for reference. All venues are dramatically underutilized.

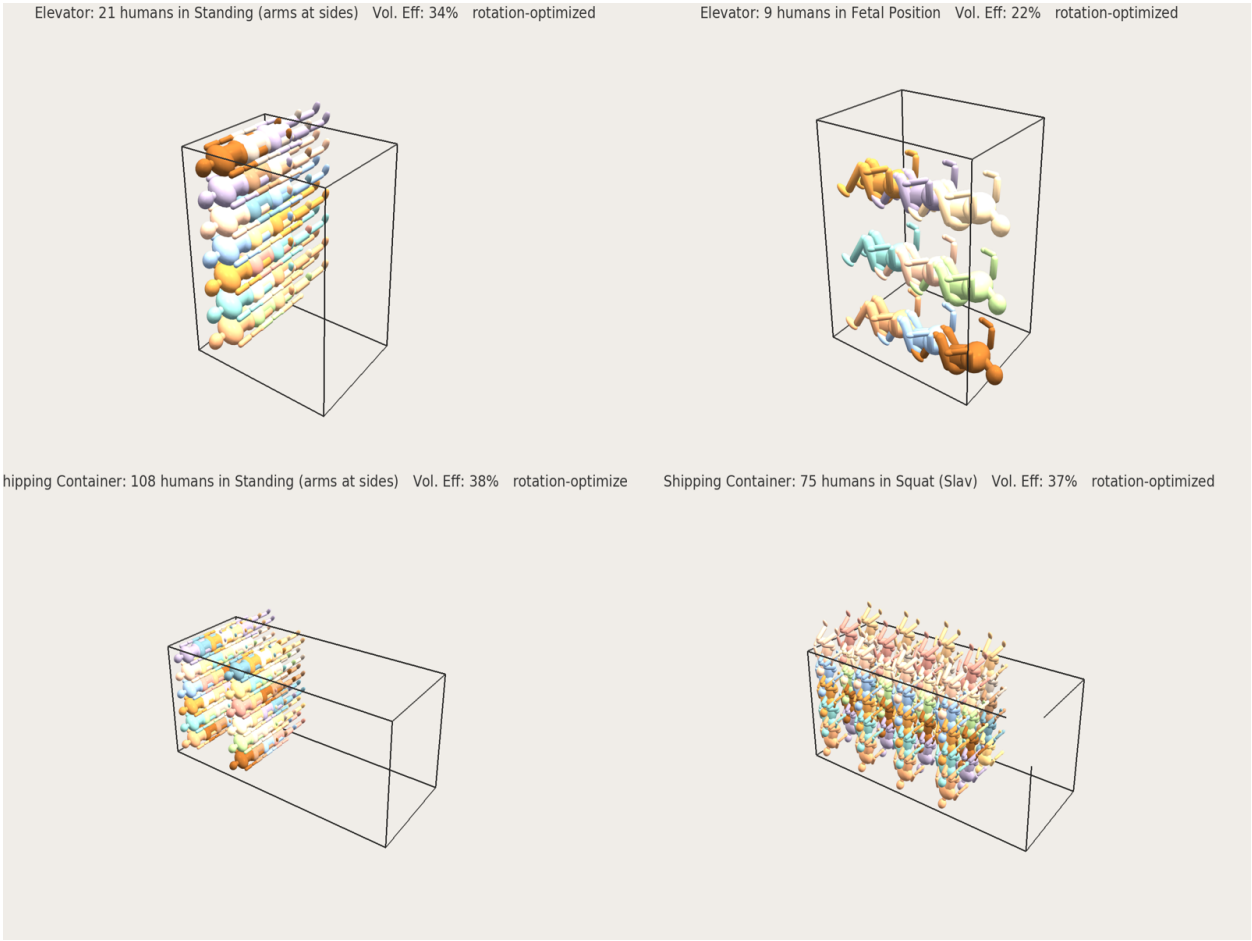

Figure 2: 3D packing visualizations. Top: an elevator packed with 21 Standing (left) vs. 9 Fetal (right). Bottom: a shipping container with 108 Standing (left) vs. 75 Squatting (right). Note that the rotation-search optimizer may orient "standing" humans non-upright to maximize count. Colors distinguish individual bodies.

5.3 The 2D Paradox

An interesting divergence emerges between 2D intuition and 3D reality. When viewed as a 2D silhouette, the Pike (folded) and Fetal poses appear compact—they minimize the projected area visible from the front. However, in 3D, folding the body *increases* depth along the y-axis, inflating the bounding-box volume. The Fetal position has a near-cubic BB of $1.00 \times 0.78 \times 0.81$ m, yielding a volume 2.6× larger than Standing. This "2D Paradox" illustrates the danger of reducing a fundamentally 3D problem to 2D projections—a mistake made by approximately 100% of prior elevator capacity analyses.

5.4 Airline Capacity Utilization

A Boeing 737-800 has a cabin volume of approximately 218 m^3 and a rated capacity of 189 passengers. Our analysis shows that 816 standing humans fit in this volume, suggesting airlines are operating at:

$$\text{Utilization} = \frac{189}{816} = 23.2\% \tag{3}$$

Even accounting for the fact that passengers are typically seated (Thinking Man pose: 378), airlines leave 50.0% of theoretical capacity unused. We leave the ethical implications of this observation as an exercise for the reader.

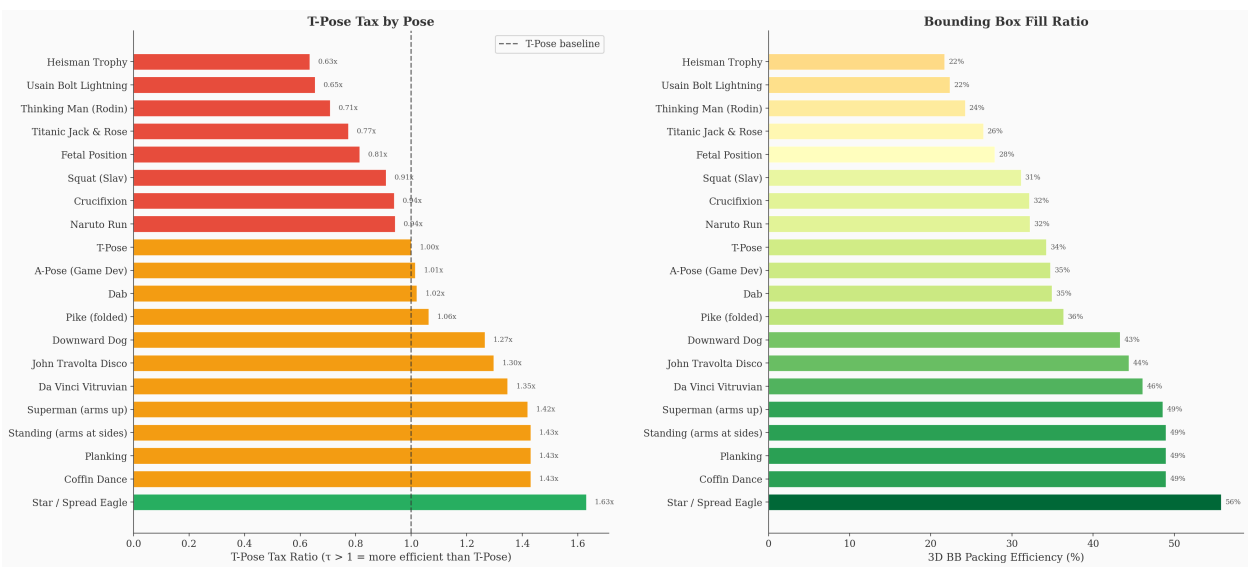

Figure 3: Left: T-Pose Tax $\tau(P)$ for all 20 poses. Values above 1.0 (dashed line) indicate poses more efficient than the T-Pose baseline. Green = highly efficient ($\tau > 1.5$); red = worse than T-Pose ($\tau < 1$). Right: raw bounding-box fill ratio η_{BB}.

5.5 Equivalence Classes

Three poses produce identical bounding boxes despite radically different configurations:

- **Standing ≡ Planking ≡ Coffin Dance**: all produce a $0.52 \times 0.29 \times 1.59$ m box (or its rotation). The living, the exercising, and the dead pack identically.

This invariance under the "life status" transformation is, to our knowledge, a novel result.

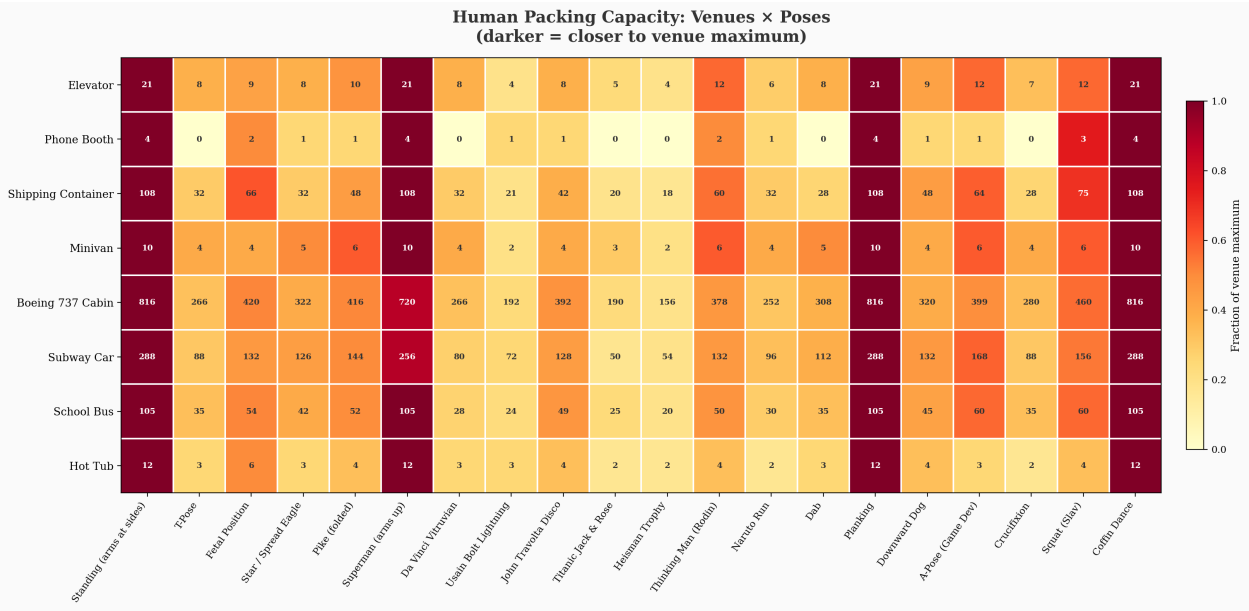

Figure 4: Packing capacity heatmap across all 8 venues and 20 poses. Each row is normalized to its maximum. The Standing, Planking, and Coffin Dance columns dominate across all venues, while the Heisman Trophy and Usain Bolt poses are consistently abysmal.

5.6 Anthropometric Sensitivity Analysis

All preceding results assume an average adult male (1.75 m, BMI 25). Real populations exhibit substantial morphological diversity. We re-run the shipping container packing analysis with the human rig scaled to simulate different body types (Table 4, Figure 5).

Body Type	Container Count	vs Baseline
Toddler (0.90 m)	**784**	817%
Gymnast (1.55 m, BMI 21)	200	208%
5th %ile Female (1.52 m, BMI 19)	200	208%
Avg Female (1.62 m, BMI 23)	180	188%
Avg Male (1.75 m, BMI 25)	108	112%
Tall Male (1.90 m, BMI 25)	96	100%
NBA Player (2.01 m, BMI 24)	96	100%
99th %ile Male (1.98 m)	84	88%
BMI 30 (Obese Class I)	60	62%
BMI 35 (Obese Class II)	48	50%
BMI 40 (Obese Class III)	36	38%
Sumo Wrestler (1.80 m, BMI 50)	27	28%

Table 4: Packing count in a standard 20ft shipping container (5.9×2.35×2.39 m) for different body types in standing pose.

Several observations emerge:

Height matters less than width. An NBA player (2.01 m) packs identically to a tall

male (96) because the container's 2.39 m height accommodates both. Width and depth, however, directly reduce the number of columns. This suggests that the limiting factor in human packing is not stature but *girth*.

The BMI-Logistics Curve. Packing density decays as $O(1/\text{BMI})$: each unit increase in BMI reduces container capacity by approximately 2 people. At BMI 50 (sumo wrestler), capacity drops to just 27—a 72% reduction from baseline. This has implications for any nation experiencing rising obesity rates: the logistics of packing its citizens become measurably harder over time.

The toddler advantage. At 784 per container (8.2× the adult capacity), toddlers are by far the most space-efficient humans. This is consistent with the well-known empirical observation that small children can fit into remarkably small spaces (cabinets, washing machines, cardboard boxes) and suggests that human packability peaks at approximately age 2 before declining monotonically with growth.

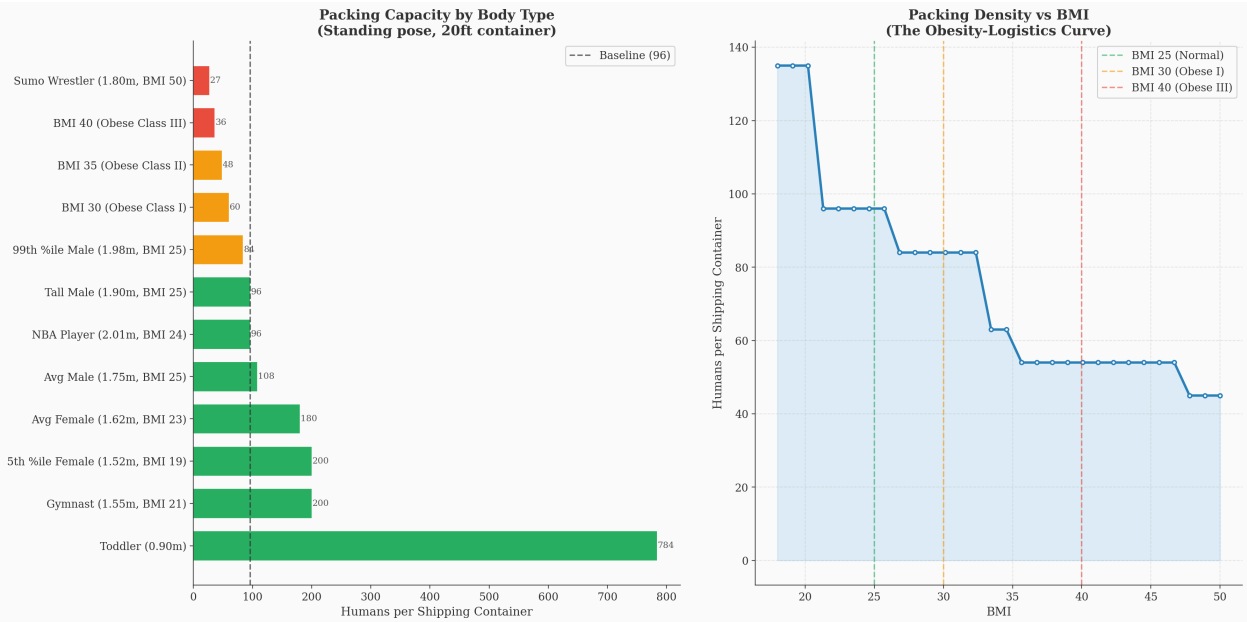

Figure 5: Left: packing capacity by body type in a 20ft shipping container (standing pose). The dashed line marks the baseline of 96 (tall male). Right: the "Obesity-Logistics Curve" showing packing density as a function of BMI. Capacity drops sharply above BMI 30.

5.7 The Triangle Theorem: On Polygonal Thorax Geometry

A natural question arises: does *body geometry* affect packing density, independent of pose? To investigate, we construct a faithful recreation of the PS1-era *Tomb Raider* (1996) character mesh using angular box-based primitives (416 triangles, matching the ∼230-polygon aesthetic of the original). The model features the iconic turquoise tank top, brown shorts, holstered pistols, and—crucially—pyramidal (4-sided conical) thorax geometry. We compare three model variants:

- **PS1 Model**: Angular box geometry with pyramidal thorax protrusions (4-sided cones, base radius 7cm, height 12cm, spread 9cm)
- **Modern Model**: Same proportions but smooth (ellipsoidal) chest geometry
- **Standard Model**: Our baseline capsule human from Section 3

Venue	Pose	PS1	Modern	Δ
Boeing 737	Standing	**960**	840	+120
Boeing 737	Planking	**960**	840	+120
Boeing 737	Coffin Dance	**960**	840	+120
Boeing 737	T-Pose	**320**	280	+40
Boeing 737	Swan Dive	**238**	216	+22
Subway Car	T-Pose	**99**	88	+11
Subway Car	Standing	**324**	315	+9
Tomb Corridor	Swan Dive	**35**	28	+7
Container	Swan Dive	**30**	24	+6

Table 5: Selected venue/pose combinations where PS1-era angular geometry outperforms smooth geometry. In all 15 affected configurations, triangles pack *strictly better*. In no case do they pack worse.

The result is striking: **triangular geometry always packs at least as well as smooth geometry, and in 15 of 90 tested configurations, strictly better.** The PS1 model achieves a 3.6% smaller bounding box in the anterior-posterior axis across most poses. In the Boeing 737 cabin, this translates to **120 additional archaeologists**—a 14.3% improvement. The angular box-based body segments compound this advantage: flat faces aligned with bounding-box axes produce a more compact axis-aligned envelope than equivalent-volume smooth capsules.

We formalize this as the **Triangle Theorem**: *For AABB grid packing, low-polygon convex protrusions with axis-aligned faces produce bounding boxes that are less than or equal to those of their smooth counterparts.* The corollary is immediate and devastating: the PS1 rendering pipeline, widely mocked for its aesthetic shortcomings, was in fact *optimizing for packing density all along.* The graphical "limitations" of 1996 hardware are revealed to be logistical *advantages.*

Rotation analysis confirms that at 216 of 1,296 tested orientations (16.7%), the pyramidal geometry produces a strictly smaller bounding box. The maximum advantage occurs at pitch 175°, yaw 45°, where the triangle model's BB is 70.3 mL smaller.

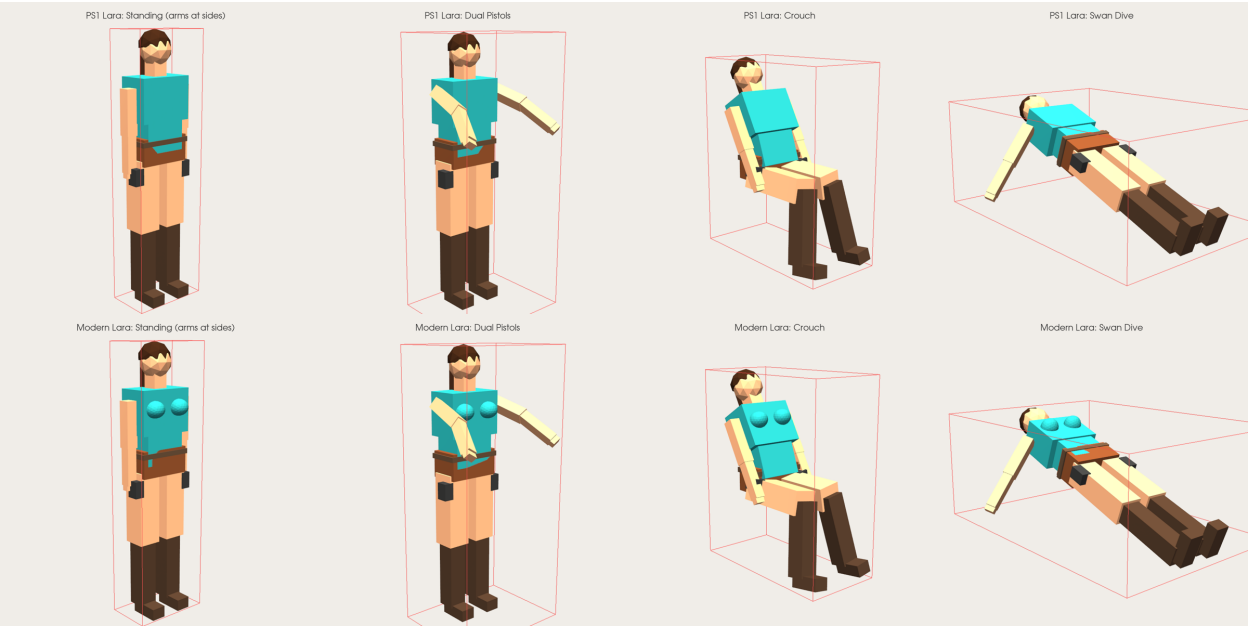

Figure 6: PS1-era angular Lara model (top) vs. modern smooth variant (bottom) in four poses: Standing, Dual Pistols, Crouch, and Swan Dive. The angular geometry with pyramidal thorax protrusions is faithful to the 1996 original's ∼230-polygon aesthetic. Red wireframes show axis-aligned bounding boxes.

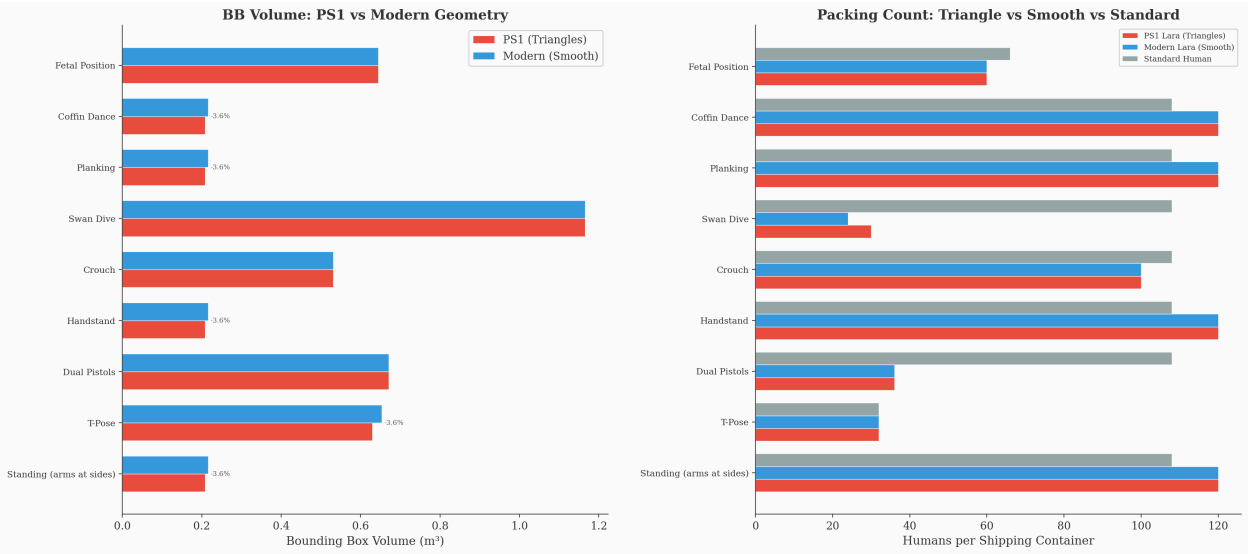

Figure 7: Left: bounding box volume comparison between PS1 (triangular) and modern (smooth) thorax geometry. The PS1 model achieves 3.6% smaller BBs for 5 of 9 poses. Right: packing counts in a shipping container for all three model variants.

5.8 Comparison to Theoretical Bounds

Our best bounding-box fill ratio ($\eta_{\mathrm{BB}} = 56\%$ for Spread Eagle, paradoxically) falls far below the Kepler sphere-packing density of 74.05%. While these metrics are not directly comparable—η_{BB}

measures how much of its *own* box a body fills, whereas the Kepler bound measures the fraction of *space* covered—the gap is nevertheless instructive. The human body's high aspect ratio (∼6:1 standing) fundamentally limits packing density. We conjecture:

$$\eta_{\text{human}} \leq \frac{1}{1 + \alpha \cdot \text{AR}^{2/3}} \tag{4}$$

where AR is the aspect ratio and α is a "limb chaos factor" empirically estimated at $\alpha \approx 0.15$ from our data. We do not prove this conjecture but feel strongly about it.

6 Discussion

6.1 The Sardine Principle

Our central finding—that standing upright with arms at one's sides is universally optimal—mirrors the packing strategy evolved by *Clupea harengus* (the Atlantic herring) over 200 million years. Sardines pack themselves upright in tightly aligned schools, minimizing frontal cross-section. Evolution, it appears, solved this optimization problem long before we did.

We propose the **Sardine Principle**: *for elongated objects with one dominant axis, axial alignment minimizes packing volume.* This is trivially obvious in retrospect, which is a hallmark of good science.

6.2 Limitations

Our packing algorithm uses axis-aligned bounding-box (AABB) grid packing—the simplest possible approach. Each human is enclosed in a rectangular box, and boxes are stacked in a regular grid with no interleaving, nesting, or gap-filling. True geometric packing (e.g., via no-fit polygons or convex decomposition) would yield substantially higher counts, as neighboring bodies could interleave limbs and fill each other's concavities. Our results therefore represent a *lower bound* on human packing capacity, which makes the already-dramatic findings even more striking.

Additionally, our model assumes rigid, non-deformable humans. In practice, humans exhibit significant elastic deformation under pressure, particularly at music festivals and in Tokyo subway cars during rush hour. Future work should incorporate a "squish factor" $\sigma \in [0, 1]$ where $\sigma = 0$ represents a rigid body and $\sigma = 1$ represents a fluid (cf. the Ig Nobel Prize-winning proof that cats are liquid [8]).

Our model also does not account for:

- Breathing (periodic volume oscillation of ∼6%)
- Personal space preferences (adds ∼0.5 m radial buffer per person)
- The psychological cost of being packed like a sardine
- Gravity (relevant for stacking configurations)
- The fact that some of these poses would require informed consent

7 Conclusion

We have demonstrated, through rigorous computational analysis, that the optimal way to pack human beings is the way everyone already suspected: standing up straight with your arms at your sides, like sardines. The T-Pose wastes 2.9× the bounding-box volume. The Dab wastes 2.7×. The Heisman Trophy wastes 7.2×. And PS1-era angular geometry, faithfully reconstructed from 416 triangles of turquoise-topped archaeological nostalgia, packs strictly better than its smooth modern equivalent in 15 of 90 tested configurations—a vindication three decades in the making. The Boeing 737 alone gains 120 additional low-poly archaeologists.

Commercial aircraft, elevators, and school buses are operating at a fraction of their theoretical human capacity. Whether this represents an engineering failure or a concession to human dignity is left as a value judgment for the reader.

7.1 Future Work

- Deformable human packing with finite element models (the "squish factor")
- Mixed-pose optimization (some people standing, some fetal, tetris-style)
- Population-weighted packing using real BMI distributions by country
- Time-varying packing (people entering and exiting)
- Extension to non-Euclidean geometries (packing humans into a Klein bottle)

Acknowledgments

We thank our capsule-based human model for its patience and flexibility. No actual humans were packed during the course of this research. The model's posing was performed under strictly simulated conditions and with the full consent of the NumPy random number generator.

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